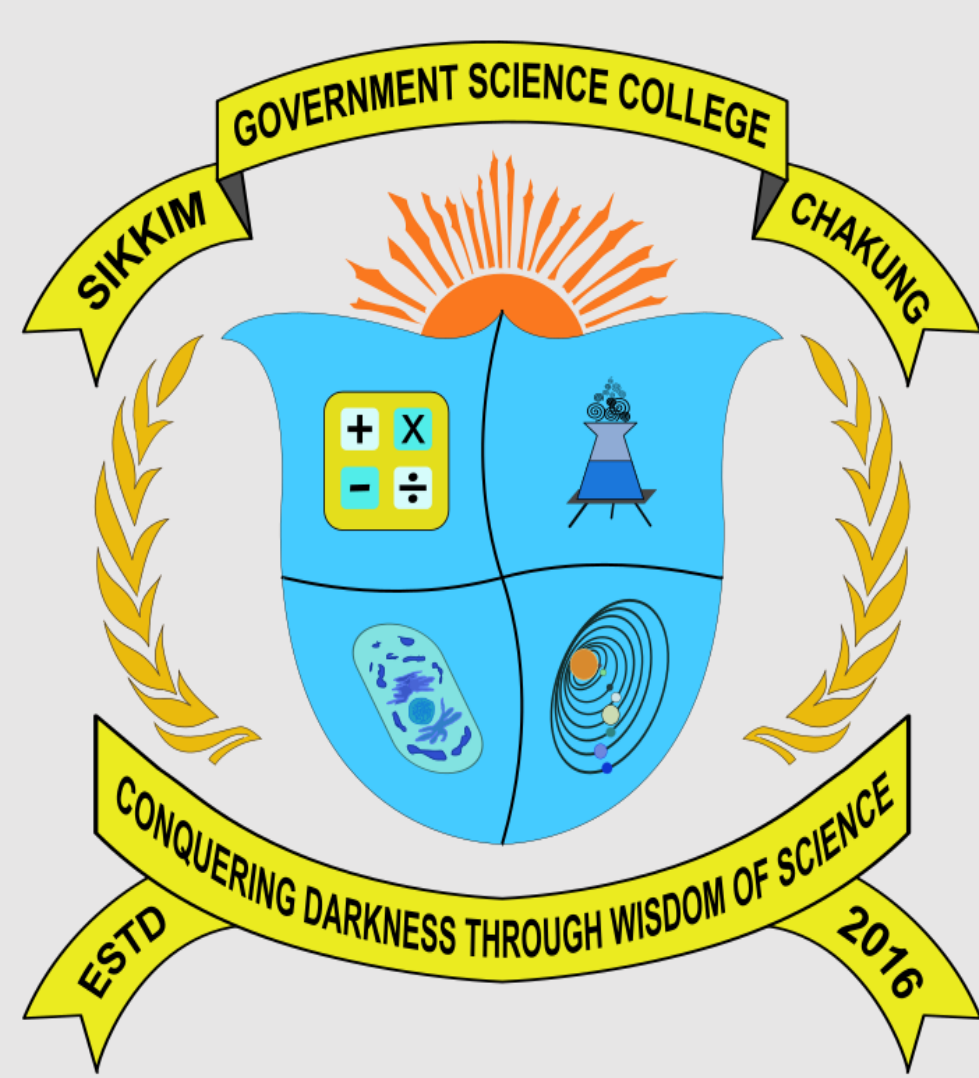




Sikkim's Shifting Climate (2009-2024): A comprehensive Review with a Physics Perspective



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Abstract

Sikkim, a mountainous state in the Eastern Himalaya, has experienced profound climate-driven changes between 2009 and 2024. This review synthesizes recent peer-reviewed studies and field reports to assess the region's evolving climate, ecological shifts, and major climate-related hazards. Key findings include consistent warming—particularly in winter—and the retreat of glaciers, resulting in the rapid expansion of glacial lakes and altered hydrology. Alpine ecosystems show significant upslope migration of plant species, while biodiversity and agricultural systems face increasing stress from temperature extremes and erratic rainfall. The most severe event during this period was the 2023 South Lhonak glacial lake outburst flood (GLOF), which destroyed critical infrastructure and displaced thousands. This and similar events underscore the growing threat of extreme weather, floods, and landslides, amplified by both climate change and unregulated development. The review highlights the urgent need for integrated risk monitoring, adaptive land use, and climate-resilient infrastructure to protect Sikkim's vulnerable communities and ecosystems in an era of accelerating environmental change.

Introduction

Sikkim, India, is a small Himalayan state in the northeast, bordered by Nepal, China, and Bhutan. Known for its mountains, biodiversity, it has a cool climate and rich forests. Located between 88° 00' 58" East and 88° 55' 25" East longitude, and 27° 04' North and 28° 07' 48" North latitude.



Figure 1: Map of Sikkim

Over the past 15 years, Sikkim has faced rapid climate-driven changes, including rising temperatures, shifting rainfall, and retreating glaciers. These changes are rooted in physical processes like increased greenhouse gas-induced warming, altered monsoon dynamics, and reduced snow and ice albedo, which accelerate glacial melt. Events like the 2023 South Lhonak Glacial Lake Outburst Flood (GLOF) reveal how these climatic shifts are intensifying natural hazards. Industrial expansion, population growth, and fragile mountain geology further amplify the risks.

This review explores how these physical and human factors are reshaping Sikkim's biodiversity, ecosystems, and communities, using evidence from scientific research, policy documents, and media sources.

Methodology

- Approach:** Integrate quantitative data with qualitative insights.
- Data Collection:** Collect historical and projected climate data from verified online meteorological and governmental archives.
- Literature Review:** Supplement data collection with a thorough review of existing research on Sikkimese climate trends and impacts.
- Local Knowledge:** Document local traditional ecological knowledge regarding climatic shifts.
- Statistical Analysis:** Apply statistical analyses (including trend analysis and anomaly detection) to the collated data.

Results and Discussion

Climate Trends

Recent studies confirm significant warming in Sikkim, with winters heating up faster than summers—mean coldest-month temperatures have risen by ~3.6 °C since the late 19th century. Modern reanalysis data show an annual warming rate of about +0.08 °C per decade, with higher rates at elevation due to elevation-dependent warming. Glaciers in the region are in negative mass balance, losing around 0.5m water equivalent per year, and retreating at accelerating rates. Precipitation patterns have also shifted, with more frequent extreme rainfall and increasingly erratic snowfall. The devastating October 2023 flood was linked to intense monsoonal downpours. A warming atmosphere—able to hold ~7% more moisture per degree Celsius—contributes to heavier rain and heightened flood risk. Snowfall is declining and shifting to higher elevations, impacting the cryosphere and water availability. Satellite imagery reveals rapidly expanding glacial lakes, such as the doubling of Kokthang Glacier's terminal lake since 1988. These changes signal a less stable climate, with serious implications for river flows, ecosystems, and disaster risk in Sikkim.

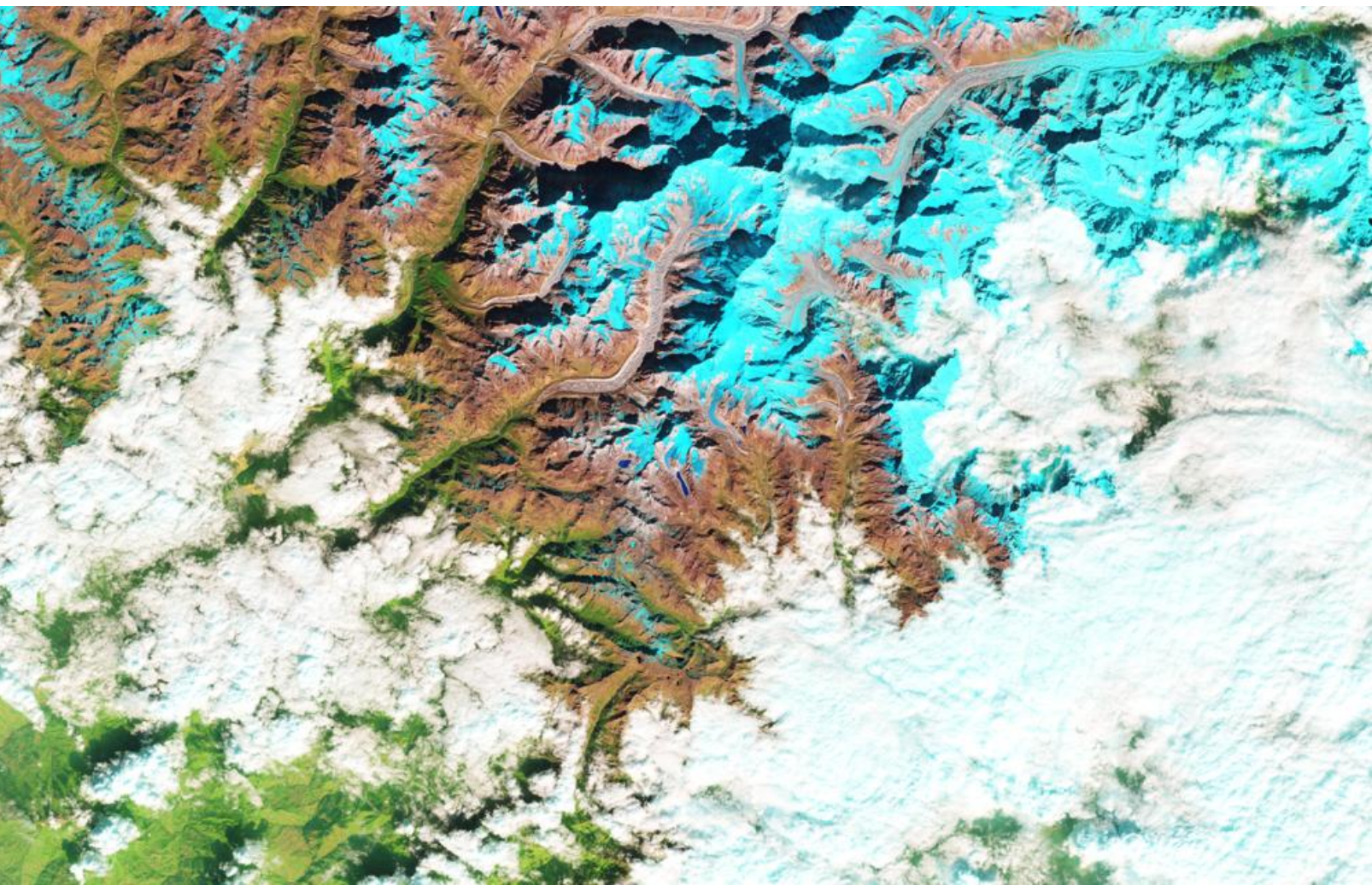


Figure 2: Glaciers retreating in Sikkim. Satellite (Landsat, Nov 2018) false-color image of the Kanchenjunga area. Snow and ice are bright cyan; vegetation is green. Glacial lakes at the foot of receding glaciers have grown substantially over recent decades. Ref: landsat.visibleearth.nasa.gov

Ecosystems and Biodiversity Changes

Sikkim's ecosystems, are undergoing climate-driven transformations. Alpine plants have shifted their elevation ranges upslope by an average of 28 m per decade, with some moving nearly 1 km higher. Animal responses include upward migration of birds, insects, and amphibians, with cold-adapted species like snow leopards at risk. Climate change also affects forests (altering growth and flowering cycles), wetlands (reducing meltwater-fed streams), and organic farming (through erratic weather). Protected area wardens report that vegetation zones are migrating upward, signaling ongoing ecological flux.

Major Climate-Related Hazards

Glacial Lake Outburst Floods (GLOFs)

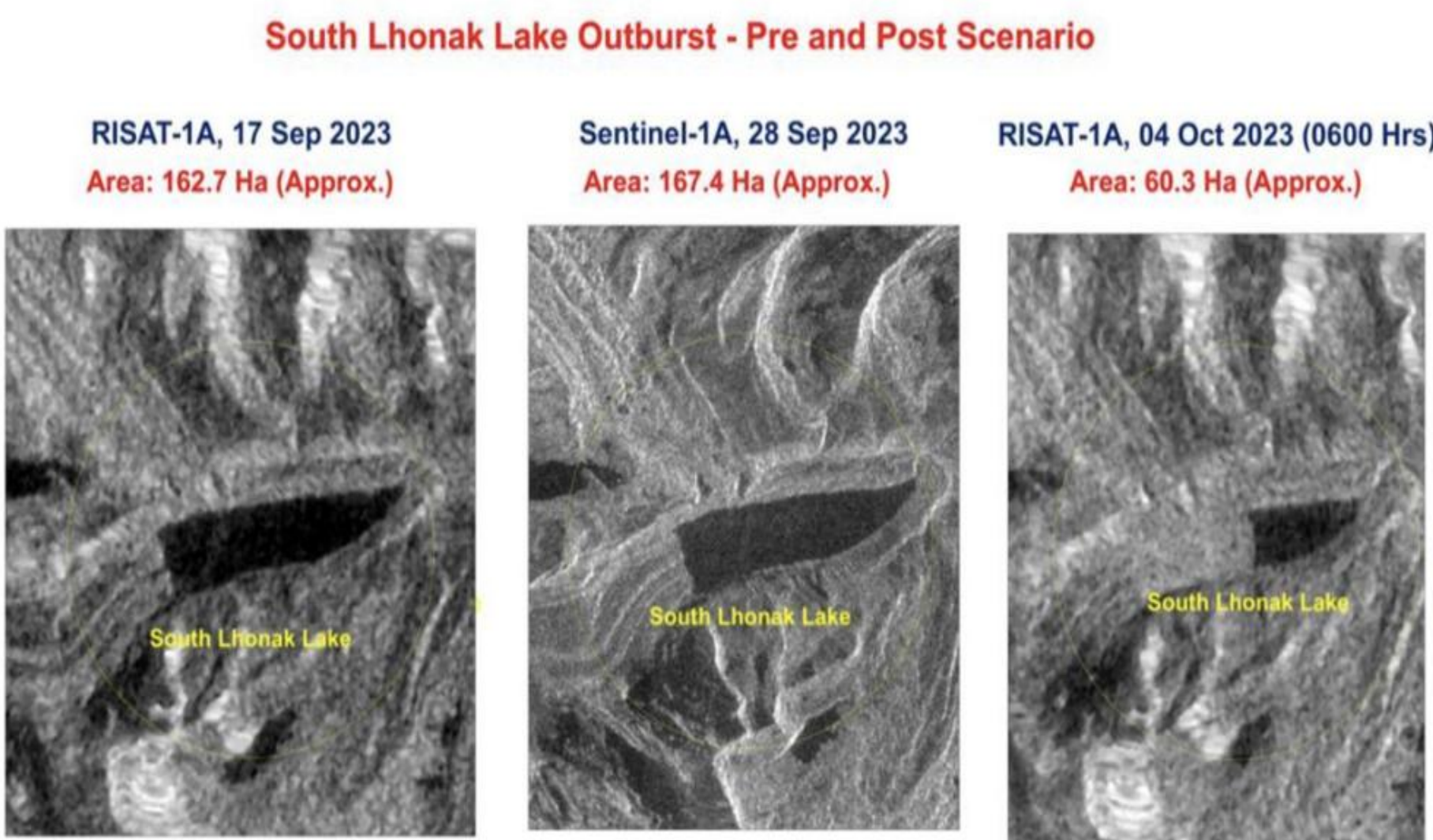


Figure 3: Pre- and post-outburst satellite images of South Lhonak Lake. SAR images from Sept 17, Sept 28 and Oct 4, 2023 show the dramatic shrinkage of the lake (labels and lake outlines in yellow). The lake's area fell from ~167 ha to ~60 ha after the dam breach. Ref: serval.unil.ch.

GLOFs are a rising threat due to glacier retreat and moraine-dammed lake expansion. The 2023 South Lhonak Lake outburst is a major example: a landslide triggered a 20 m-high wave that released 50 million m³ of water, destroying the Teesta-III dam and killing at least 55 people. Physics shows that moraine collapse and high lake volume caused extreme discharge (up to 5,340 m³/s). Warming trends and unstable slopes make such hazards increasingly likely, with dozens of high-risk glacial lakes still under surveillance.

Floods and Rainfall Extremes



Figure 4: Flooding in the Teesta River, resulting from glacial outbursts and erratic rainfall, is causing landslides and road destruction

Heavy monsoon rains have intensified, triggering frequent flash floods and landslides. Events in 2022 and 2023 brought deadly floods and record-breaking rainfall, especially in Teesta Valley. Saturated soils from intense downpours amplify GLOF and landslide risks. Climate models predict more such events as warmer air holds more moisture, increasing the potential for destructive cloudbursts.

Landslides and Slope Failures

Landslides are common due to Sikkim's steep terrain, but climate change adds risk through heavier rainfall and thawing permafrost. Construction activity also weakens slopes. Recent examples like the August 2024 Teesta Stage V landslide show how even moderate or delayed rains can destabilize hillsides, especially where prior damage exists. The convergence of rainfall, geology, and human activity demands better slope monitoring and early warning systems.

Human Development and Climate Impacts

Population growth, infrastructure development, and tourism are reshaping Sikkim. While hydropower and roads support economic growth, they also increase vulnerability to climate hazards. Dam failures (like Teesta-III in 2023) and unplanned construction on unstable slopes are key concerns. Tourism adds pressure on fragile ecosystems, while agriculture faces challenges from temperature and rainfall shifts, despite Sikkim's 100% organic status. Sikkim's Climate Action Plan promotes sustainability, but balancing growth with environmental safety remains difficult. As development accelerates, the risks from climate extremes grow – requiring integrated planning, conservation, and adaptive infrastructure.

Physics insight on Climate Impacts in Sikkim

Though largely conceptual, basic physics helps explain Sikkim's climate-related changes:

Thermodynamics of snow and ice: Glacier melt requires latent heat. Warmer air and increased solar radiation provide more energy, accelerating melting. Each °C of warming increases longwave radiation and sensible heat flux. As glaciers retreat, they lose reflective snow cover (albedo), exposing darker surfaces that absorb more heat—creating a feedback loop of further melt.

Atmospheric changes: The Clausius–Clapeyron equation,

$$\frac{d}{dt}e_s = \frac{L_v e_s}{R_v T^2}$$

describes how saturation vapor pressure (e_s) increases exponentially with temperature. Practically, air holds ~7% more moisture per °C warming, intensifying rainfall. This was seen in Sikkim's 2023 flood: warmer temperatures and Bay of Bengal moisture triggered record rain. Rising temperatures also shift precipitation from snow to rain, reducing slow snowmelt and increasing rapid runoff.

Hydrodynamics of GLOFs: The 2023 GLOF released ~50 million m³ of water at ~10 m/s, yielding kinetic energy of ~2.5×10¹² J—enough to destroy major infrastructure. A collapsing moraine (~14.7 million m³) displaced lake water, triggering a ~20 m wave. Such waves conserve energy in narrow valleys, overwhelming downstream barriers like the Teesta-III dam, ~60 km away.

Permafrost feedbacks: Warming thaws permafrost in high moraines. This phase change consumes latent heat but greatly reduces soil cohesion. Even minor thawing can destabilize slopes. In the South Lhonak case, permafrost degradation likely played a decisive role in triggering the outburst.

Conclusion

Over the past 15 years, Sikkim has become a vivid example of how climate change impacts fragile mountain systems. Rising temperatures, erratic monsoons, and rapid glacier retreat—intensified by local factors like albedo loss and terrain instability—have led to biodiversity loss, frequent landslides, floods, and disasters like the 2023 South Lhonak GLOF.

At the same time, industrial growth, hydropower projects, and expanding settlements have outpaced the region's ecological and infrastructural resilience. Despite warnings, gaps persist in climate risk management, early warning systems, and environmental regulation.

Safeguarding Sikkim's ecological and cultural heritage demands urgent, climate-resilient infrastructure, stricter land-use policies, and adaptation strategies that blend traditional knowledge with science. Sikkim shows that Himalayan climate risks are no longer future threats—they are already here and demand immediate, integrated action.

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